Available online at www.sciencedirect.com

ScienceDirect

Advances in Space Research xxx (xxxx) xxx

ADVANCES IN SPACE RESEARCH (a COSPAR publication)

www.elsevier.com/locate/asr

A reinforcement learning-based cooperative distributed evolutionary algorithm for the electromagnetic detection satellite scheduling

Yanjie Song [a], Yangyang Guo [b], Lining Xing [c], Jian Wu [d], Jixuan Wang [e,*], Shuaiheng Huai [a], Witold Pedrycz [f,g,h]

[a] School of Information Science and Technology, Dalian Maritime University, Dalian, China
[b] School of Systems Science, Beijing Jiaotong University, Beijing, China
[c] Research Institute of Intelligent Control and Manufacturing System, Jiangsu University of Technology, Changzhou, China
[d] Hangzhou Institute of Technology, Xidian University, Hangzhou, China
[e] School of Artificial Intelligence, Hebei University of Technology, Tianjin, China
[f] Silesian University of Technology (SUT), Department of Measurement and Control Systems, Gliwice, Poland
[g] Department of Electrical and Computer Engineering, University of Alberta, Edmonton, Canada
[h] Constructor University, Bremen, Germany

Received 7 April 2026; received in revised form 25 May 2026; accepted 6 June 2026

Abstract

Electromagnetic detection satellites (EDSs) can accurately capture changes in the electromagnetic environment in various areas, including the air, ground, sea, and others, by acquiring electromagnetic signals through onboard equipment. To fully meet user needs within limited satellite capacity and number constraints, it is crucial for satellite control units to schedule EDSs efficiently. In this regard, we propose a reinforcement learning-based cooperative distributed evolutionary algorithm (RL-CDEA) to address the EDS scheduling problem (EDSSP). The RL-CDEA incorporates multiple subpopulations for global exploration and local exploitation respectively. Furthermore, a reinforcement learning-based individual transfer mechanism dynamically adjusts the composition of these subpopulations. Additionally, RL-CDEA employs a diversified crossover strategy. Each subpopulation responsible for spatial exploration adopts different evolutionary approaches to identify high-quality task execution schemes. Extensive simulation experiments demonstrate that our proposed algorithm effectively solves EDSSP while outperforming comparison algorithms across several aspects. This study highlights the efficacy of a cooperative distributed search model with an information interaction mechanism in addressing complex combinatorial optimization problems such as the EDSSP.
© 2026 COSPAR. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Evolutionary algorithm; Cooperative distributed; Reinforcement learning; Electromagnetic detection satellites; Scheduling

1. Introduction

Capturing electromagnetic signals on Earth and in space is essential for understanding changes in the electromagnetic environment and safeguarding national security

* Corresponding author.
E-mail addresses: songyj_2017@163.com (Y. Song), 24121704@bjtu.edu.cn (Y. Guo), xln@jsut.edu.cn (L. Xing), jianwwwu@126.com (J. Wu), wangjixuan@hebut.edu.cn (J. Wang), huaishuaiheng@dlmu.edu.cn (S. Huai), wpedrycz@ualberta.ca (W. Pedrycz).

https://doi.org/10.1016/j.asr.2026.06.023
0273-1177/© 2026 COSPAR. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Please cite this article as: Y. Song, Y. Guo, L. Xing et al., A reinforcement learning-based cooperative distributed evolutionary algorithm for the electromagnetic detection satellite scheduling, Advances in Space Research, https://doi.org/10.1016/j.asr.2026.06.023

Table 1
Variables and symbols.

Symbol	Description
Sat	set of electromagnetic detection satellite resource
sat_i	electromagnetic detection satellite i
T	set of detection tasks
$task_j$	detection task j
ept_j	earliest permissible start time for detection task j
lpt_j	latest permissible end time for detection task j
dur_j	length of detection time required for detection task j
p_j	the profit of successful implementation of detection task j
G_0	initial signal gain
φ	angle between antenna and target
λ	wavelength
D	antenna diameter
η	power factor of the antenna
m_j	memory occupied by detection task j
e_j	energy consumed by detection task j
$\widehat{O}_i$	orbital set of satellite i
O_{ij}	set of orbits visible to task j from satellite i
o_{ijk}	visible orbit k of the electromagnetic detection satellite i for task j
tw_{ijk}	time window of satellite i in orbit k for task j
evt_{ijk}	earliest visible time of satellite i in orbit k for task j
lvt_{ijk}	latest visible time of satellite i in orbit k for task j
M_i	memory capacity of satellite i
E_i	energy storage limit of satellite i
Δ_i	transition time for satellite i to execute new task
I	a large positive integer
x_{ijk}	whether satellite i executes task j in orbit k If the task is executed, $x_{ijk}=1$, else $x_{ijk}=0$
st_{ijk}	start time of satellite i to execute task j in orbit k

2.3. Mathematical model

Based on Song et al. (2023b, 2014), the assumptions of the EDSSP are as follows:

1. All detection tasks are known before the scheduling process begins. During the scheduling horizon, no new tasks are added, no existing tasks are removed, and the task requirements, including permissible execution time ranges, remain unchanged.

2. Each satellite can receive and execute the assigned detection tasks according to the generated scheduling plan.

3. All satellites remain in normal operational condition throughout the scheduling horizon. Satellite platform failures, payload malfunctions, and other unexpected equipment anomalies are not considered in this model.

4. At the beginning of each orbital revolution, each satellite is recharged through its solar panels and is assumed to have sufficient energy storage for subsequent scheduling decisions.

5. Before each orbital revolution begins, the onboard storage of each satellite is cleared, so the full storage capacity is available for the tasks scheduled in the subsequent orbital revolution.

The optimization objective of our study is to identify a task execution scheme that maximizes detection profit. This objective function setting aims to fully meet the user's practical usage requirements while efficiently using satellite resources. The specific form of the objective function and constraints are presented below.

Objective Function:

$$\max \sum_{i\in Sat}\sum_{j\in T}\sum_{k\in O_{ij}} p_j \cdot x_{ijk} \tag{1}$$

Referring to Ref. Song et al. (2023b), the profit of task can be calculated by Eq. 2-Eq. 4. It can be seen that the smaller the angle between the satellite's antenna and the target area the better the quality of the received signal. The profit obtained from accomplishing the task is also higher. The calculation process of the detection profit is:

$$p_j = G_0 \cdot \left[\frac{J_1(u)}{2u} + 36\frac{J_3(u)}{u^3}\right]^2 (dBi) \tag{2}$$

$$G_0 = \eta\frac{\pi^2 D^2}{\lambda^2}(dBi) \tag{3}$$

where $u = 2.07123\sin(\varphi)/\sin(\varphi_{3dB}), J_1(u)$ and $J_3(u)$ are the 1st and 3rd order Bessel functions of the first kind, respectively. φ is the angle between the satellite antenna and the center of the signal source, φ_{3dB} is the angle at which the antenna gain is attenuated by $3dB$ relative to the center of the beam, and the calculation formula as follows.

$$\varphi_{3dB} = 70\lambda/D \tag{4}$$

where λrepresents the wavelength, and D indicates the diameter of the antenna.

Constraint:

$$\sum_{k\in\widehat{O}_i} m_j \cdot x_{ijk} \leqslant M_i, \forall i \in Sat, j \in T \tag{5}$$

$$\sum_{k\in\widehat{O}_i} e_j \cdot x_{ijk} \leqslant E_i, \forall i \in Sat, j \in T \tag{6}$$

$$\sum_{j\in T}\sum_{k\in O_{ij}} m_j \cdot x_{ijk} - \left(\left|O_{ij}\right| - 1\right)\cdot M_i \leqslant M_i, \forall i \in Sat \tag{7}$$

$$\sum_{j\in T}\sum_{k\in O_{ij}} e_j \cdot x_{ijk} - \left(\left|O_{ij}\right| - 1\right)\cdot E_i \leqslant E_i, \forall i \in Sat \tag{8}$$

$$ept_j \cdot x_{ijk} \leqslant st_{ijk}, \forall i \in Sat, j \in T, k \in O_{ij} \tag{9}$$

$$\left(st_{ijk} + dur_j\right)\cdot x_{ijk} \leqslant lpt_j, \forall i \in Sat, j \in T, k \in O_{ij} \tag{10}$$

$$evt_{ijk} \cdot x_{ijk} \leqslant st_{ijk}, \forall i \in Sat, j \in T, k \in O_{ij} \tag{11}$$

$$\left(st_{ijk} + dur_j\right)\cdot x_{ijk} \leqslant lvt_{ijk}, \forall i \in Sat, j \in T, k \in O_{ij} \tag{12}$$

$$\left(st_{ijk} + dur_j\right)\cdot x_{ijk} + \Delta_i \leqslant st_{ijk\prime} + I\cdot\left(1 - x_{ijk\prime}\right), \forall i \in Sat, j, j\prime \in T, k, k\prime \in O_{ij} \tag{13}$$

$$\sum_{i\in Sat}\sum_{k\in O_{ij}} x_{ijk} \leqslant 1, \forall j \in T \tag{14}$$

$$x_{ijk} \in \{0, 1\} \tag{15}$$

The constraint 5 indicates that the satellite's fixed storage memory within a single orbit cannot exceed the upper limit of the total fixed storage memory. Constraint 6 indicates that the energy consumption of the satellite within a single orbit cannot exceed the upper limit of the energy storage.

Constraint 7 indicates that the storage consumption over the entire scheduling period cannot exceed the satellite's upper limit. Constraint 8 indicates that energy consumption over the entire scheduling period cannot exceed the satellite's energy limit. Constraint 9 indicates that tasks need to start after the earliest permissible execution time. Constraint 10 indicates that tasks need to be completed before the latest permissible execution time. Constraint 11 and 12 indicate that tasks need to be executed within the time windows. Constraint 13 indicates that one satellite needs to fulfill the transition time requirement between two tasks. After executing one task, the satellite needs to take some time to reconfigure its parameters before it can start executing the next task. Here, the Big M method is used to ensure that the inequality relation holds. The transition time Δ is a constant, which is determined according to the satellite's payload capacity. The value of this time is communicated to the satellite control unit by the payload producer when the satellite is in use. The satellite management and control unit also monitors whether the conversion time changes during the daily use of the satellite. If there is a change, the transition time is also updated in a timely manner. Constraint 14 indicates that each task can be executed at most once. Constraint 15 indicates the range of values for the decision variable.

3. The proposed approach

To address the EDSSP, we propose an enhanced EA framework and employ a RL-assisted algorithmic search approach, referred to as RL-CDEA. The overall algorithm flow and key steps are elaborated in detail within this section.

3.1. Overall framework of RL-CDEA

The complexity of EDSSP makes the algorithm solution difficult. According to the problem characteristics, we improve the EA to obtain RL-CDEA, which has three obvious advantages over the traditional EA: 1) This algorithm uses three subpopulations to perform cooperative distributed search, and the subpopulations are optimized by diversified population evolution; 2) Two global search methods and a local search method designed according to the problem are used for the three subpopulations, which find a high-profit solution for the detection task execution as much as possible; 3) An RL-based individual migration mechanism is used to dynamically adjust the composition of individuals within the subpopulations to effectively improve the efficiency of the search. For the convenience of expressing the three subpopulations, the first subpopulation is denoted as P_1, the second subpopulation is denoted as P_2, and the third subpopulation is denoted as P_3. These subpopulations collaborate extensively to explore the problem space comprehensively. After each generation of search is completed, the individual migration mechanism completes the migration of individuals between subpopulations based on the current state (i.e., the algorithmic search situation). The overall flow of the proposed algorithm is shown in Fig. 2.

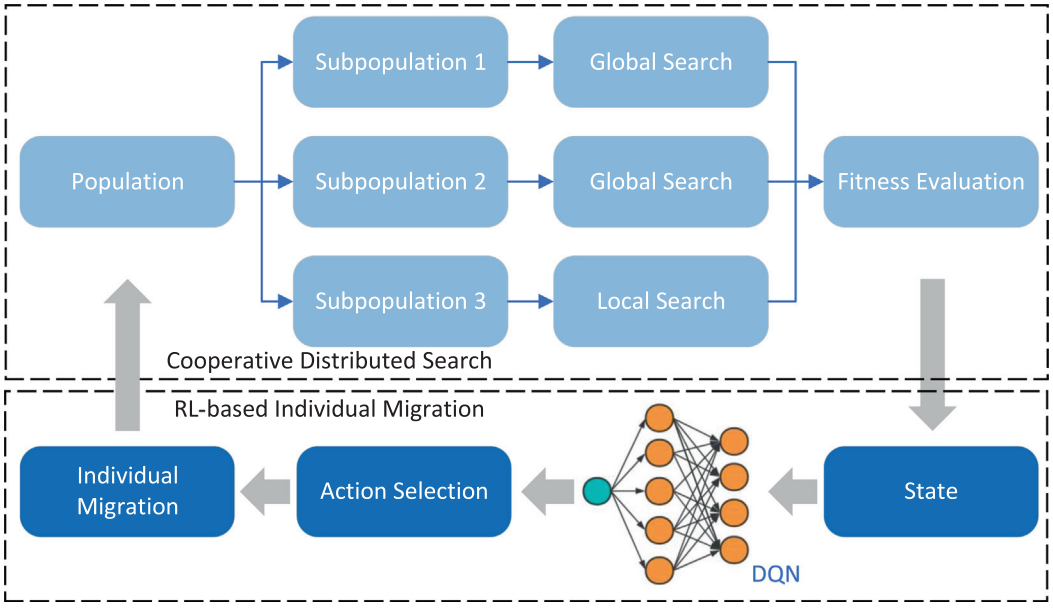


Fig. 2. Overall flow of the RL-CDEA.

Algorithm 1. The RL-CDEA

Input: Sat,O,T,TW,DQN,N_p^1,N_p^2,N_p^3, threshold for updating weights $Thre_1$, threshold for updating subpopulation $Thre_2$,L, control parameter for crossover α, control parameter for mutation β,MFE
Output: best individual glo_best, task execution scheme obtained by decoding the best individual $Solution$

1 Initialize algorithm parameters;
2 Set time step $\tau \leftarrow 0$, fitness evaluation times $fet \leftarrow 0$;
3 $P_0 \leftarrow$ Generate initial population using a randomized approach;
4 $glo_best, glo_indi \leftarrow$ Evaluate the fitness function value of the population;
5 **while** fet *is smaller than* MFE **do**
6 //Subpopulation Cooperative Search
7 Execute adaptive crossover for subpopulation (P_1, α);
8 Execute two-point crossover for subpopulation (P_2, α);
9 Execute local search for subpopulation P_3 by Algorithm 3;
10 Execute mutation in each subpopulation (P_1, P_2, P_3, β);
11 $loc_best, loc_indi \leftarrow$ Evaluate fitness value of the population;
12 Update operators' scores based on search performance by Algorithm 2;
13 $fet \leftarrow$ Update the fitness evaluation times;
14 **if** $loc_best > glo_best$ **then**
15 $glo_best \leftarrow loc_best$;
16 $glo_indi \leftarrow loc_indi$;
17 **if** $mod\,(fet, Thre_1) == 0$ **then**
18 Update operators' weights;
19 **if** $mod\,(fet, Thre_2) == 0$ **then**
20 Update subpopulation P_3;
21 $r_\tau \leftarrow$ Compute reward by Eq. 16;
22 Record state transition;
23 $a_\tau \leftarrow$ Select action using DQN;
24 $(P_1', P_2', P_3',) \leftarrow$ Move individuals according to the action(P_1, P_2, P_3, a_τ);
25 $\tau \leftarrow \tau + 1$;
26 $last_best \leftarrow loc_best$ //Store the best individual of the current population;

The pseudo-code of RL-CDEA is presented in Algorithm 1. Following random initialization, the subpopulations carry out optimization based on their respective predefined search methods (Lines 6–9). Subsequently, the operator's score is updated after each generation of population search (Line 11). If the conditions for weight update are met, the operator's weight values are recalculated (Line 18–20).

3.2. Encoding and decoding

The processes of encoding and decoding serve as the fundamental components of EA optimization and solution acquisition. In RL-CDEA, we employ integer encoding (i.e., task sequence number) for our approach. To illustrate the encoding process in detail, let us consider a chromosome sequence with a length of 5. As depicted in Fig. 3, the initial encoded value "3" within the chromosome segment represents the 3-rd detection task from the task set. The second gene value '5' indicates the 5-th probe in the set of tasks. Similarly, subsequent encoded values "1", "4", and "2" correspond to the 1st, 4th, and 2nd tasks respectively.

The evolution of RL-CDEA all takes place on chromosomes with integer encoding composition. This algorithm sequentially arranges tasks during the decoding process based on the order of chromosome encoding, aiming to determine whether a task can be successfully scheduled according to the constraints specified in Section 2.3. If all the constraints are satisfied during the attempted arrangement, the task can be executed successfully; otherwise, it needs to be abandoned for scheduling. Tasks are attempted to be arranged one by one in the order in which they appear in the chromosome. The task arrangement method is referenced from Song et al. (2023b). Specifically, when a task is given, an attempt is first made to schedule it at the position with the smallest angle between the receiver facing and the target position. If this position does not satisfy all of the constraints, then an attempt will be made to move the position backwards in execution start time. The task selects the earliest position at which it can start executing the task as it moves backward. If the task keeps moving backwards until the task completion time has exceeded the latest visible time in the time window, the task will be abandoned to try to schedule the next task. The total profit obtained from decoding represents the fitness value of an individual.

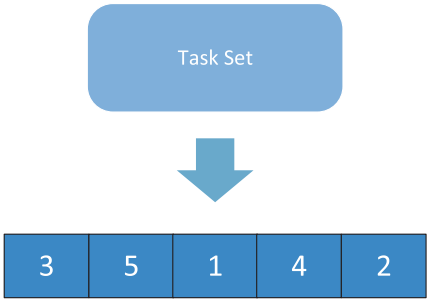


Fig. 3. An example of encoding.

3.3. Division of subpopulations

RL-CDEA employs a cooperative distributed search approach to identify high-profit solutions for detection task execution. The algorithm consists of three subpopulations that effectively explore the problem solution space at both global and local levels. The first two subpopulations (P_1 and P_2) primarily focus on global exploration, while the last subpopulation (P_3) is responsible for local exploitation. The size of the three subpopulations is N_p^1, N_p^2, and N_p^3, respectively. The composition of these subpopulations and the specific optimization operations performed are described in detail below.

The initial generation of the first subpopulation P_1 and the second subpopulation P_2 is achieved through randomization at the onset of the algorithmic search. Subsequent iterative searches will be conducted following a similar approach as described in Section 3.4 for diverse crossover and Section 3.5 for mutation operation. The overarching objective of this global search effort, encompassing multiple styles of subpopulations, is to identify the detection scheme that maximizes detection profit.

The third subpopulation, denoted as P_3, comprises individuals with the highest fitness values in their respective subpopulations of P_1 and P_2. This subpopulation conducts a comprehensive local search, which is elaborated upon in Section 3.6. It should be noted that the update of subpopulation P_3 occurs after a predetermined number of algorithmic searches have been executed. Furthermore, for the construction of P_3, one optimal individual from each subpopulation of P_1 and P_2 will be re-selected.

3.4. Diverse crossover

Based on the multiple subpopulations introduced in the previous section, we propose a diverse crossover strategy (DCS) employed in the population search process of P_1 and P_2. The DCS uses distinct crossovers for different subpopulations, namely adaptive crossover and simple crossover. Adaptive crossovers possess the capability to dynamically adjust the utilized operators based on algorithmic search, while simple crossovers aim to comprehensively explore the entire solution space through random search. The astute combination of these two types of crossovers enables RL-CDEA to thoroughly explore the solution space and facilitate the identification of global optimal solutions. Subsequently, this section provides detailed descriptions of these two categories of crossovers.

The subpopulation P_1 employs adaptive crossover to efficiently explore the solution space of the problem. Adaptive crossover encompasses a range of operators that facilitate differential exploration. Among these, the primary five operators include two-point crossover, multi-point crossover, flip, reconstruction, and heuristic adjustment. The subsequent section provides a detailed description of the specific implementation of these five operators.

Two-point Crossover: Two chromosome segments, each with a length of L, are selected from an individual. A new chromosome is obtained by exchanging the positions of the two fragments, as illustrated in Fig. 4.

Multi-point Crossover: Two sets of genes of total length L each are selected from within an individual. By swapping positions according to the depicted process in Fig. 5, a new chromosome is generated without altering the internal relative order.

Flip: A chromosome fragment of length L is selected from an individual. The fragments are depicted in Fig. 6, where they are rearranged in reverse order to generate a novel chromosome.

Reconstruction: A chromosome segment of length $2L$ is selected from within an individual. The genes within this fragment are randomly disrupted and reinserted into their original positions to form a complete chromosome. The reconstruction operation is shown schematically in Fig. 7.

Heuristic adjustment: A chromosome segment of length L is selected from within an individual, and the genes within this segment are adjusted in ascending order based on the earliest allowable execution time of the task. Subsequently, the adjusted chromosome segment is reinserted into its original position to form a complete chromosome.

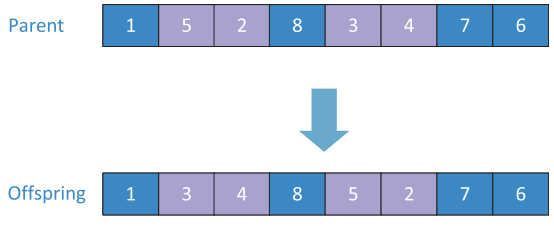


Fig. 4. Two-point Crossover. Chromosome segments in purple execute crossover. Chromosome segments in blue do not change during crossover.

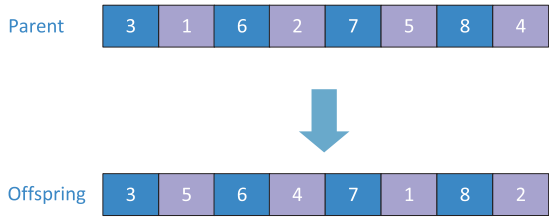


Fig. 5. Multi-point crossover.

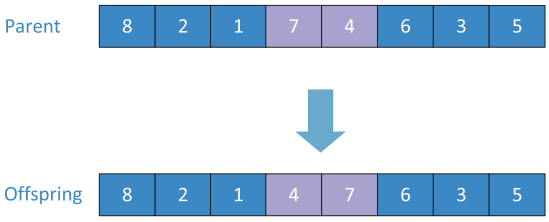


Fig. 6. Flip.

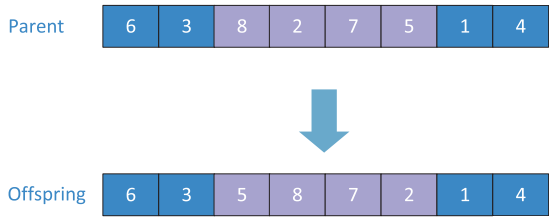


Fig. 7. Reconstruction.

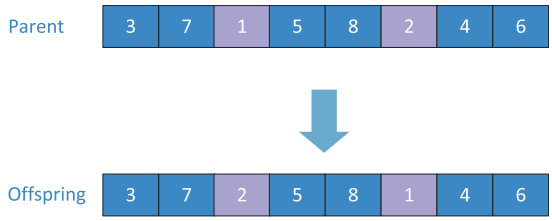


Fig. 8. Mutation.

The adaptive crossover initializes equal scores for each operator at the start of the algorithm's search, ensuring an equal probability of selection. The corresponding weights can be easily calculated based on the operator's score relative to all scores, enabling scientific operator selection using the roulette method. As operators are used in an iterative search, their respective scores are updated according to optimization performance.

The score update is categorized into four cases. If the population fitness function value obtained from the search is superior to the optimal value discovered in the previous search, then the score will be updated to $score_1$. If it is inferior to the previous search's optimal value but better than that of the prior generation's population search, then it will be updated to $score_2$. In case it is worse than both previous searches' optimal values but within an acceptable range, then it will be updated to $score_3$. Finally, if it is worse than only the most recent best value but still within an acceptable range μ compared with that of prior generations' population searches, then its score will be updated as $score_4$. Algorithm 2 presents a pseudo-code for updating the score.

Algorithm 2. Update score

Input: $last_best$,loc_best,glo_best,μ,$score_1$,$score_2$,$score_3$, $score_4$,selected operator score $score_s$
Output: updated operator scores $score'_s$

```
if loc_best > glo_best then
    score'_s ← score_s + score_1;
if loc_best ≤ glo_best and loc_best > last_best then
    score'_s ← score_s + score_2;
if loc_best ≤ last_best and loc_best > last_best ∗ μ then
    score'_s ← score_s + score_3;
if loc_best ≤ last_best ∗ μ then
    score'_s ← score_s + score_4;
```

The crossover operator continuously updates its score dynamically throughout the iterative search process based on the comparison results of fitness values. After several updates, the operator score is recalculated to determine the probability of selection (i.e., weight value). The specific calculation method remains consistent with that used during initialization.

The subpopulation P_2 crossover is simplified, employing solely the previously described two-point crossover technique. The utilization of this approach offers the advantage of maximizing search randomness, thereby facilitating the discovery of novel chromosome structures that exhibit significant dissimilarity from previously explored regions.

In addition, subpopulation P_3 will not use crossover and mutation, but will use local search for solution updating. The specific details regarding the implementation of local search will be expounded upon in Section 3.6.

3.5. Mutation

The algorithm employs both crossover and mutation for exploitation during population search to enhance its performance. The mutation is uniformly applied within each subpopulation of RL-CDEA, following a simple procedure. Specifically, an individual undergoes mutation by randomly selecting two genes from its chromosome and exchanging their positions to generate a new chromosome

sequence (refer to Fig. 8 for the schematic representation of this operation).

3.6. Local search

The RL-CDEA employs a population-based approach for global exploration and incorporates individual local search for exploitation. The pseudo-code for the local search is presented in Algorithm 3. In the first step of the local search, an individual (Lines 2–3) is selected to perform subsequent operations. To ensure selection bias towards dominant individuals while allowing relatively poorer individuals some possibility of being chosen, we employ a 0–1 random number to select from two individuals based on their fitness values relative to the total fitness values of the population.

Algorithm 3. Local Search

Input: subpopulation P_3,task set T, fitness of subpopulation F
Output: updated subpopulationP_3',fitness evaluation times for local search fet_{LS}

1 Set $fet_{LS} \leftarrow 0$;
2 Calculate the proportion of individual fitness values within the P_3 to the total fitness value of P;
3 $indi, f \leftarrow$ Select an individual using roulette method(P_3, F);
4 $loc_1 \leftarrow$ Selects a random location other than $|T|$ as the starting position for the search;
5 **while** $loc_1 < |T|$ **do**
6 Set $loc_2 \leftarrow loc_1 + 1$;
7 $indi' \leftarrow$ Exchange gene in the individual$(indi, loc_1, loc_2)$;
8 $f' \leftarrow$ Evaluate fitness value of the new individual$(indi')$;
9 $fet_{LS} \leftarrow fet_{LS} + 1$;
10 **if** $f' > f$ **then**
11 $indi \leftarrow indi'$;
12 $f \leftarrow f'$;
13 $loc_1 \leftarrow loc_1 + 1$;
14 **else**
15 Loop while;
16 **if** $loc_1 == |T|$ **then**
17 $loc_1 \leftarrow loc_1 + 1$;
18 **Return**: updated subpopulation P_3', fitness evaluation times for local search fet_{LS}

After completing the individual selection, the algorithm updates the individual by iteratively swapping gene positions (Lines 6–7). Specifically, a gene is randomly selected from within the individual's chromosome at one position and exchanged with the gene at the next position. If this exchange leads to an improvement in the fitness function value, further swaps are performed between genes at adjacent positions until no further improvement is observed, terminating the search operation.

3.7. RL-based individual migration mechanism

The RL-CDEA efficiently optimizes based on the subpopulations divided by Section 3.3. To enhance its search capability, we propose an RL-based individual migration mechanism. This mechanism allows individuals within multiple subpopulations to dynamically adjust their behavior according to search performance. This section describes in detail the basis for the construction of the mechanism, the construction process, and the related settings.

The prerequisite for combining RL with EA is to satisfy the conditions necessary for constructing a Markov decision process (MDP). In RL-CDEA, which aims to solve the EDSSP, multiple subpopulations are explored based solely on the current state of the population and the adopted evolutionary strategy. The optimization process is independent of both previous and subsequent population compositions. This characterization of population optimization precisely fulfills the prerequisites for MDP construction, enabling us to define relevant properties in the RL method according to the problem at hand.

The recent advancements in RL have led to the development of a range of innovative methods based on value, strategy, and value-strategy combinations. In the case of EDSSP, we utilize the classical DQN method for implementing dynamic optimization of agents. The DQN can be defined by its attributes $\langle S, A, R, TR \rangle$, where S represents state, A represents action, R represents reward, and TR represents state transition. In the following sections, we will provide detailed explanations for each attribute's specific meaning within the context of this algorithm.

State: The three subpopulations (P_1, P_2, P_3) and their corresponding fitness function values are combined as a unified input to the RL model.

We use the combination of fitness values from the subpopulation and the flattened subpopulation as input to the DQN. Fig. 9 illustrates the vector composed of states. When constructing the vector, the encoding values of pop-

ulation and fitness function values are first normalized separately.

Action: The action set A contains a total of seven individual migration actions. These individual migration actions are described as follows.

a_1: Select an individual from P_1 randomly, and then select an individual from P_2 randomly to exchange their location with each other.

a_2: Select an individual from P_1 randomly, and then select an individual from P_3 randomly to exchange their locations with each other.

a_3: Select an individual from P_2 randomly, and then select an individual from P_3 randomly to swap locations with each other.

a_4: Select the best individual in the population from P_1, and then select the best individual in the population from P_2 to exchange their positions with each other.

a_5: Select the best individual in the population from P_1, and then select the best individual in the population from P_3 to exchange their positions.

a_6: Select the best individual in the population from P_2, and then select the best individual in the population from P_3 to exchange their positions with each other.

a_7: No individual migration operation is performed.

The seven migration actions are designed to provide a compact yet representative action space for the DQN. Actions a_1–a_3 randomly exchange individuals between each pair of subpopulations, which promotes population diversity and cross-population exploration. Actions a_4–a_6 exchange the best individuals between each pair of subpopulations, enabling high-quality scheduling information to be shared and accelerating exploitation. Action a_7 performs no migration, allowing the DQN to avoid unnecessary exchanges when the current population structure is already effective. Therefore, the action set balances exploration, exploitation, and search stability while keeping the output dimension of the DQN manageable.

Reward: The change in the value of the optimal fitness function for the population search is defined as the reward. The reward value is calculated as shown in Eq. 16.

$$r_\tau = f^*_{local} - f^*_{last} \tag{16}$$

where r_τ denotes reward, f^*_{local} denotes the best fitness function value of the current population, and f^*_{last} denotes the best fitness function value of the previous generation population.

The value function plays a crucial role in all RL methods, including DQN Fan et al. (2020). In the RL framework, the agent adjusts the parameter configurations of the network model based on environmental feedback to seek an optimal policy Ladosz et al. (2022). This policy is computed as depicted in Eq. 17.

$$y_\tau = r_\tau + \gamma Q\left(s_{\tau+1}, \max_{a_{\tau+1}} Q(s_{\tau+1}, a_{\tau+1}; \theta); \theta\right) \tag{17}$$

where y_τ denotes the target Q-value, Q denotes the action-value function, γ denotes the discount factor.

The optimal value function represents the objective we aim to discover, necessitating the training of network model parameters θ through backpropagation to approach an ideal state as closely as possible. During the backpropagation algorithm, the loss function can be formulated as follows:

$$L_i(\theta_i) = \mathbb{E}_{s,a\sim\rho(\cdot)}\left[(y_i - Q(s,a;\theta_i))^2\right] \tag{18}$$

where $\mathbb{E}$ denotes the mathematical expectation and ρ denotes the distribution of data sampled from the relay buffer.

If the model parameters are replicated to the action-value function network using model parameter replication, it becomes easier to compute the loss values Sutton and Barto (2018). Based on the above description, we can pre-

Algorithm 4. Training algorithm of DQN

Input: Relay Buffer,DQN,time step τ,step interval SI,batch size B_s,learning rate α_L,discount factor γ.
Output: updated action-value function Q network model parameters θ

```
if mod (t, SI) == 0 then
    Q̂ ← Replicate Q-network model parameters;
B ← Extract B_s amount of data from Relay Buffer;
for i = 1 to B_s do
    PV ←select data from batch B in order to input into the network model Q̂(φ'_i, a'; θ̂);
    if s_{τ+1} is not a termination state then
        y_i = r_i + γ max_{a'}(PV);
    else
        y = r_τ
Compute the loss function using (Y_i − Q(φ_i, A_i; θ))^2;
θ ← Optimize model parameters by gradient descent using an optimizer(α_L, γ);
```

sent the DQN training method as depicted in Algorithm 4. An action-value function network and a target-value function network will be used for model training. The network model training employs a relay buffer mechanism that records and dynamically updates the usage of the model. In the Algorithm 4, the function ϕ represents the conversion of state S into a fixed-length data format.

4. Experiments

The quality, optimization speed, and effectiveness of the enhanced strategy in the proposed algorithm for solving the EDSSP and generating task execution plans are thoroughly validated through simulation experiments.

4.1. Experimental settings

Experimental Platform: All the experiments are conducted on a desktop computer with a CPU of Core I7-12700, 32 GB of RAM, and an operating system version of Windows 11 23H2. The coding environment is Python 3.9.7.

Experimental Instances: Since EDSSP currently has no public dataset for algorithm performance experiments, we use random generation to obtain test instances. The parameters of the randomly generated instances are configured as follows: The time range of the task is within one day, the number of detection tasks is 100–1000, and the required detection time is included in the range of 10–120s. Following the above parameter settings, we generate a number of randomized test instances on the actual EDS control system. In order to represent the different instances, we adopt the form of "Task Scale-No." for representation. "Task Scale" denotes the scale of the instance. "No." denotes the specific number of the instance.

Comparison Algorithms: To validate the advantages of RL-CDEA for solving EDSSP, three state-of-the-art algorithms for effectively solving EDSSP with other types of exploration satellite scheduling problems are chosen as comparison algorithms. These three algorithms are: reinforcement learning-based memtic algorithm (RL-MA) Song et al. (2023a), reinforcement learning-based genetic algorithm (RL-GA) Song et al. (2023b), and improved adaptive large neighborhood search algorithm (ALNS-I) Liu et al. (2024). RL-MA employs the RL method to automate the selection of local search operators integrated into the memetic algorithm. RL-GA uses an RL-based self-adaptive operator selection mechanism to optimize the algorithm search. ALNS-I designs a series of destruction and repair operations based on the characteristics of the satellite scheduling problem.

Parameter Configuration: The algorithm parameters for RL-CDEA are configured as follows. The maximum number of fitness evaluations is set to 5000, the population size of the algorithm is set to 10, the size of N_P^1 is set to 4, the size of N_P^2 is set to 4, the size of N_P^3 is set to 2, the crossover length L is set to 2, the crossover probability is set to 0.95, the mutation probability is set to 0.05, $Thre_1$ is set to 100, $Thre_2$ is set to 200, acceptance percentage μ is set to 0.95, interval step is set to 10, batch size is set to 32, number of hidden layers is set to 4, discount factor is set to 0.9, the learning rate is set to 0.0001, the optimizer uses Stochastic Gradient Descent (SGD) Optimizer, loss function uses Mean Squared Error (MSE) and the score is set to 40, 30, 20, 10 respectively. The parameter settings of the other algorithms remain consistent with those in the relevant literature.

Evaluation Criteria: In order to fairly compare the solution performance of these stochastic optimization algorithms, each algorithm will be run independently 30 times under each size of instances. The algorithms will be statistically evaluated for their optimized maximum (denoted as Max) and mean (denoted as Mean) at the end of the run. In addition, the Wilcoxon rank sum test (denoted as WD) is also used to evaluate the difference between the algorithms, which is denoted by +/-/= to indicate that the comparison algorithm outperforms/equals/inferior to the proposed algorithm, respectively.

4.2. Analysis and discussion

At first, we conducted a comprehensive statistical and analytical analysis of the profit achieved by the algorithmic search-based execution schemes for the detection task. As presented in Table 2, our proposed RL-CDEA outperforms the three compared algorithms significantly in terms of both optimal values obtained from the search process and average performance. This substantial difference can be attributed to two main factors: firstly, RL-CDEA employs diverse search strategies that enhance its effectiveness; secondly, it incorporates an information exchange mechanism between sub-populations. Particularly for complex problems like EDSSP, relying solely on population search does not always obtain high-performing detection schemes. The combination of global and local searches proves more effective when dealing with larger task scales as supported by our experimental results. Notably, as the task scale increases further, the advantages offered by our proposed algorithm become even more pronounced.

Subsequently, the search stability and convergence of the algorithm were verified. As shown in Fig. 10, RL-CDEA has less fluctuation between each search result. This is mainly related to the way the algorithm balances global and local search. When the problem size is large, the local search plays an indispensable role in allowing the algorithm to find solutions with similar optimal performance. In the comparison algorithms, RL-MA also performs better in terms of overall stability due to the presence of local search. However, the ability of RL-MA to balance global and local search is not as good as RL-CDEA. The lack of local search in the remaining two comparison algorithms does not guarantee that the results of each search will be stable within a small interval.

Table 2
Comparison of algorithms' optimization results. The bolded font indicates the best result among the separate statistical indicators.

Instance	RL-CDEA		RL-MA		RL-GA		ALNS-I	
	Max	Mean	Max	Mean(WD)	Max	Mean(WD)	Max	Mean(WD)
100–1	**1288**	**1288.00**	**1288**	**1288.00=**	**1288**	**1288.00=**	**1288**	**1288.00=**
100–2	**1210**	**1210.00**	**1210**	**1210.00=**	**1210**	**1210.00=**	**1210**	**1210.00=**
100–3	**1225**	**1225.00**	**1225**	**1225.00=**	**1225**	**1225.00=**	**1225**	**1225.00=**
200–1	**2304**	**2304.00**	**2304**	**2304.00=**	**2304**	**2304.00=**	**2304**	**2304.00=**
200–2	**2520**	**2520.00**	**2520**	**2520.00=**	**2520**	2518.87-	**2520**	2514.07-
200–3	**2330**	**2330.00**	**2330**	**2330.00=**	**2330**	**2330.00=**	**2330**	**2330.00=**
300–1	**3797**	**3788.47**	3775	3757.53-	3774	3755.93-	3759	3730.87-
300–2	**3727**	**3704.60**	3670	3653.20-	3683	3662.13-	3651	3623.47-
300–3	**3953**	**3903.27**	3878	3860.53-	3888	3868.67-	3874	3833.60-
400–1	**4922**	**4922.00**	**4922**	4920.87-	**4922**	**4922.00=**	**4922**	4903.60-
400–2	**4694**	**4694.00**	**4694**	**4694.00=**	**4694**	**4694.00=**	**4694**	4693.67-
400–3	**5054**	**5037.67**	5029	5003.87-	5019	5005.33-	5005	4970.27-
500–1	**6224**	**6202.07**	6186	6159.33-	6186	6168.00-	6171	6136.67-
500–2	**6194**	**6180.00**	6169	6150.13-	6174	6158.20-	6169	6117.00-
500–3	**5897**	**5879.27**	5861	5834.93-	5867	5839.87-	5835	5793.07-
600–1	**7224**	**7163.07**	7066	7031.27-	7077	7042.87-	7029	6970.80-
600–2	**7274**	**7217.40**	7161	7126.73-	7175	7131.20-	7127	7081.47-
600–3	**7269**	**7221.87**	7098	7058.80-	7140	7085.60-	7061	6989.67-
700–1	**8754**	**8729.87**	8679	8653.60-	8695	8662.13-	8664	8593.53-
700–2	**8280**	**8276.20**	8275	8257.07-	8275	8261.27-	8262	8224.13-
700–3	**8468**	**8447.53**	8447	8432.33-	8453	8436.27-	8435	8391.93-
800–1	**9477**	**9448.47**	9393	9308.27-	9372	9327.60-	9342	9260.87-
800–2	**9901**	**9865.20**	9792	9747.00-	9804	9776.20-	9786	9700.07-
800–3	**9512**	**9438.20**	9342	9312.53-	9367	9324.20-	9322	9266.33-
900–1	**11241**	**11192.20**	10976	10927.47-	11046	10964.53-	10921	10842.20-
900–2	**11036**	**10939.13**	10760	10674.13-	10741	10690.47-	10669	10592.73-
900–3	**10590**	**10523.07**	10382	10306.33-	10374	10309.33-	10338	10196.07-
1000–1	**11316**	**11248.80**	11013	10951.47-	11014	10950.20-	10929	10866.47-
1000–2	**11341**	**11278.80**	11069	10974.00-	11058	10971.93-	10944	10863.60-
1000–3	**11716**	**11590.53**	11416	11304.13-	11372	11318.33-	11252	11173.27-
+/−/=		–		0/23/7		0/23/7		0/25/5

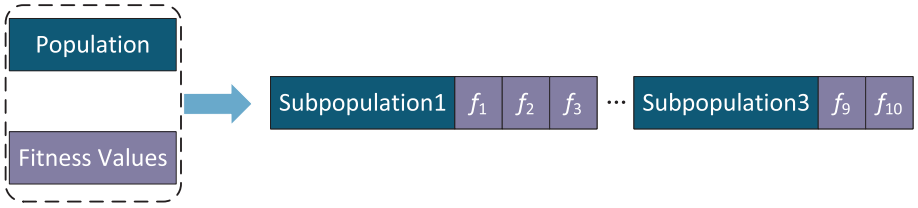
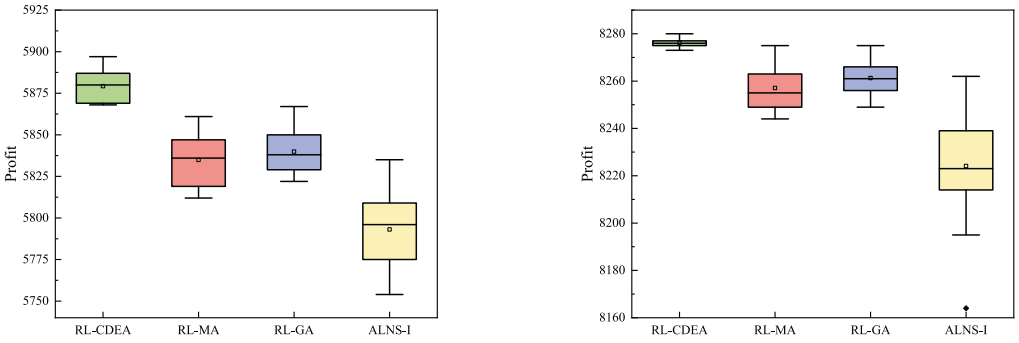


Fig. 9. Input vector for DQN.

As shown in Fig. 11, RL-CDEA converges faster after initialization and obtains higher detection profits than the comparison algorithms. This advantage becomes more evident as the task scale increases, mainly because the cooperative distributed search and inter-subpopulation information exchange help balance global exploration and local exploitation.

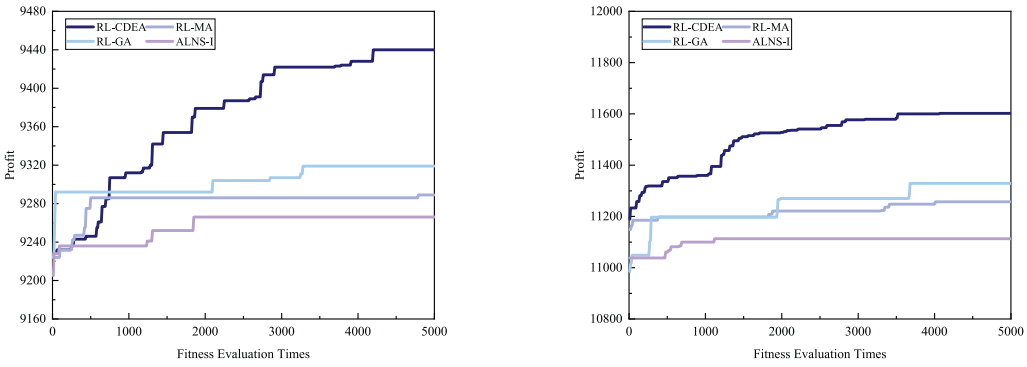
The performance of the algorithms in terms of search efficiency was also analyzed. As evident from the results presented in Table 3, RL-CDEA exhibits superior search speed compared to the comparison algorithms. The absence of heuristic initialization in RL-CDEA significantly saves search time. The algorithm also has a low time cost during the search process, which also ensures that the algorithm takes less time to search in each iteration. In particular, the core RL-based individual migration mechanism requires a bite-sized amount of time for each action selection, which is considerably lower than the time cost associated with elite repository and individual update strategies employed by the comparison algorithms.

Furthermore, we conducted algorithm comparison experiments to validate the efficacy of the enhanced strategies employed in our algorithm. Specifically, we individually removed certain strategies from RL-CDEA and denoted them as RL-CDEA-W1 (without diversity crossover) and RL-CDEA-W2 (without RL-based individual



(a) Comparison on the Stability of 500 Task Scale (b) Comparison on the Stability of 700 Task Scale

Fig. 10. Comparison on the stability of algorithms.



(a) Convergence Performance at 800 Task Scale (b) Convergence Performance at 1000 Task Scale

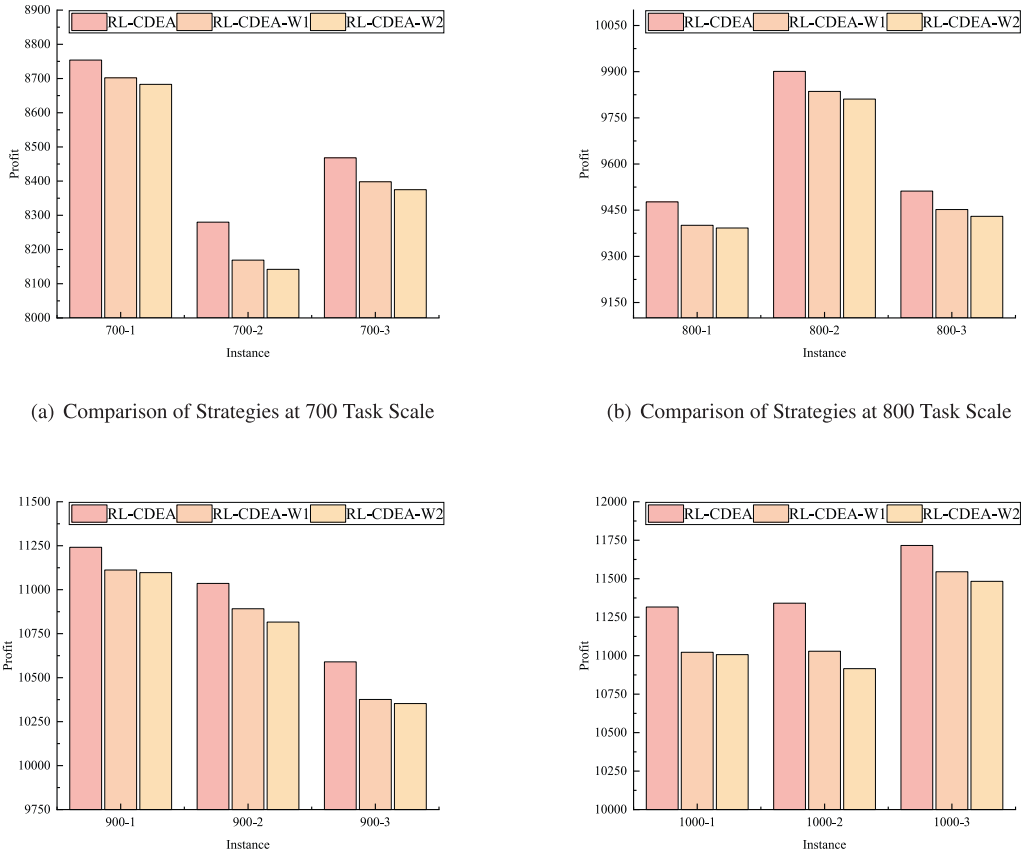
Fig. 11. Comparison on the convergence performance of algorithms.

Table 3
Results of average running CPU time.

Instance	RL-CDEA	RL-MA	RL-GA	ALNS-I
100	**50.31**	59.42	53.16	60.29
200	**101.35**	123.87	110.04	125.73
300	**152.40**	171.76	163.19	172.12
400	**316.92**	348.25	337.30	352.36
500	**450.27**	482.19	478.22	489.51
600	**543.16**	584.93	579.15	603.24
700	**875.58**	925.31	912.70	942.16
800	**963.53**	1034.28	1024.58	1047.30
900	**1149.83**	1206.29	1195.61	1223.47
1000	**1248.64**	1317.82	1309.25	1326.85

migration mechanism). The experimental results are presented in Fig. 12, clearly demonstrating that the utilization of these improved strategies significantly enhances the search performance of our algorithm. Among all the enhancement strategies, the RL-based individual migration mechanism contributes most to improving population search efficiency. This can be attributed to the information exchange between sub-populations facilitated by individual migration, which enables effective knowledge sharing and ultimately boosts search effectiveness.

We conducted a sensitivity analysis of key parameters in the algorithm. The two parameters—subpopulation composition and crossover length, were examined. We set up three configurations with different subpopulation sizes. Configuration 1 is N_p^1=4, N_p^2=4, N_p^3=2. Configuration 2 is N_p^1=4, N_p^2=2, N_p^3=4. Configuration 3 is N_p^1=5, N_p^2=4, N_p^3=1. The crossover length is set to 2, 4, and 6, respectively. As shown in Fig. 13-(a), the different subpopulation sizes affect the quality of the solutions obtained. For RL-



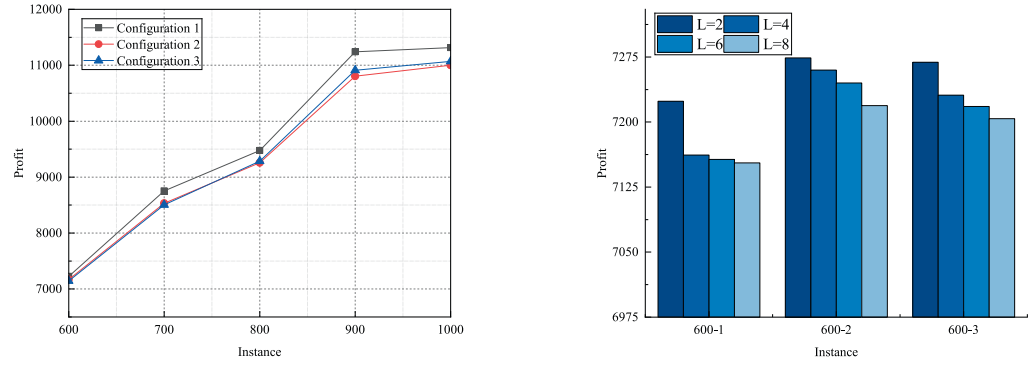
(a) Comparison of Strategies at 700 Task Scale

(b) Comparison of Strategies at 800 Task Scale

(c) Comparison of Strategies at 900 Task Scale

(d) Comparison of Strategies at 1000 Task Scale

Fig. 12. Validation of the effectiveness of algorithmic strategies.



(a) Parameter Sensitivity Analysis of Subpopulation Composition

(b) Parameter Sensitivity Analysis of Crossover Length

Fig. 13. Parameter sensitivity analysis.

CDEA, population-based search plays a critical role. If the search focuses primarily on neighborhood search, the algorithm's performance cannot be optimized. Fig. 13-(b) illustrates the impact of different crossover fragment lengths on the algorithm's search performance. It can be observed that excessively long crossover fragment lengths do not enable the algorithm to find higher-quality solutions. An appropriate crossover fragment length helps the algorithm optimize problems with large solution spaces, such as EDSSP.

The extensive experimental results above demonstrate the efficient solving capability of the proposed RL-CDEA for EDSSP. This algorithm not only exhibits a significant advantage in terms of total detection profit obtained from the search but also surpasses comparison algorithms in search stability and time efficiency. Particularly, its performance superiority on large-scale problems enables RL-CDEA to be effectively applied in real scheduling scenarios. The algorithm's diversity of searches and cooperative sub-

population search pattern enable it to effectively handle various complex situations.

It is worth noting that this work represents a commendable endeavor towards implementing a cooperative distributed evolutionary algorithm. The demonstrated performance benefits resulting from the utilization of cooperative distributed search warrants further in-depth investigation. Additionally, the RL approach employed in the algorithm enhances search performance singularly, while advanced AI techniques can optimize the overall algorithmic performance through various means. Naturally, it is crucial to acknowledge and strike a balance between algorithmic performance and elevated computational overhead during the design process.

5. Conclusion

This study introduces a novel solution approach for the EDSSP, wherein a new scheduling model is devised based on the problem at hand. In our study, we have taken into account two additional factors, memory consumption and power consumption, compared to other related studies. This makes the study more relevant to practical applications. The model considers satellite normal operation and mission execution, while also utilizing problem characteristics to guide the scheduling algorithm's design. RL-CDEA, which contains multiple subpopulations, achieves a good balance between global and local search by means of cooperative distributed search. By leveraging the inherent strengths of EA and incorporating an RL approach, RL-CDEA achieves higher average detection profits than the three baseline algorithms, namely RL-MA, RL-GA, and ALNS-I. RL-CDEA improves the scheduling performance by approximately 1.5%, 1.4%, and 2.2% over RL-MA, RL-GA, and ALNS-I, respectively.

In future research, we will consider more specific scenarios, such as satellite failures and temporary mission changes, requiring the design of specialized algorithms according to the problem's characteristics. Moreover, the algorithm design will also incorporate other advanced artificial intelligence methods, like large language models, which will enable the algorithm to have more powerful learning and search capabilities.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (72501042, 72501214), Key Laboratory of Intelligent Space TTC&O (CYK2024-02–04).

References

Berger, J., Lo, N., Barkaoui, M., 2020. QUEST: a new quadratic decision model for the multi-satellite scheduling problem. Comput. Oper. Res. 115, 104822.

Bu, H.-J., Zhang, J., Luo, Y.-Z., 2016. Constraint satisfaction and optimization for space station short-term mission planning based on an iterative conflict-repair method. Eng. Optim. 48 (10), 1658–1678.

Chen, X., Reinelt, G., Dai, G., Spitz, A., 2019. A mixed integer linear programming model for multi-satellite scheduling. Eur. J. Oper. Res. 275 (2), 694–707.

Du, Y., Wang, T., Xin, B., Wang, L., Chen, Y., Xing, L., 2020. A data-driven parallel scheduling approach for multiple agile Earth observation satellites. IEEE Trans. Evol. Comput. 24 (4), 679–693.

Shi, Z.E.R., Gan, L., Baoyin, H., Li, J., 2021. Multi-satellites imaging scheduling using individual reconfiguration based integer coding genetic algorithm. Acta Astronaut. 178, 645–657.

Fan, J., Wang, Z., Xie, Y., Yang, Z., 2020. A theoretical analysis of deep Q-learning. Proc. Mach. Learn. Res. 120, 486–489.

Kilic, S., Ozkan, O., 2022. Modeling and optimization approaches for satellite broadcast scheduling problem. IEEE Trans. Eng. Manage. 69 (5), 1848–1864.

Kim, H., Chang, Y.-K., 2020. Optimal mission scheduling for hybrid synthetic aperture radar satellite constellation based on weighting factors. Aerosp. Sci. Technol. 107, 106287.

Kisseleff, S., Lagunas, E., Abdu, T.S., Chatzinotas, S., Ottersten, B., 2020. Radio resource management techniques for multibeam satellite systems. IEEE Commun. Lett. 25 (8), 2448–2452.

Ladosz, P., Weng, L., Kim, M., Oh, H., 2022. Exploration in deep reinforcement learning: a survey. Inform. Fusion 85, 1–22.

Li, Z., Li, X., 2019. A multi-objective binary-encoding differential evolution algorithm for proactive scheduling of agile Earth observation satellites. Adv. Space Res. 63 (10), 3258–3269.

Li, J., Chen, H., Zhong, Z., Jing, N., Wu, J., 2014. Method for electromagnetic detection satellites scheduling based on genetic algorithm with alterable penalty coefficient. J. Syst. Eng. Electron. 25 (5), 822–832.

Liu, Z., Liu, J., Liu, X., Yang, W., Wu, J., Chen, Y., 2024. Knowledge-assisted adaptive large neighbourhood search algorithm for the satellite–ground link scheduling problem. Comput. Industr. Eng. 192, 110219.

Qu, Q., Liu, K., Li, X., Zhou, Y., Lü, J., 2023. Satellite observation and data-transmission scheduling using imitation learning based on mixed integer linear programming. IEEE Trans. Aerosp. Electron. Syst. 59 (2), 1989–2001.

Ren, Y., Yuxin, Q., Yang, D., Ruifeng, X., Qianxu, N., Keyi, Z., Song, Y., Zhang, Y., 2026. A survey on preference-based multi-objective evolutionary algorithms. Mathematics 14 (8), 1365.

Sarkheyli, A., Vaghei, B.G., Bagheri, A., 2010. New tabu search heuristic in scheduling Earth observation satellites. In: Proceedings of the 2010 2nd International Conference on Software Technology and Engineering, vol. 2, V2–199–V2–203.

Shao, Y., Lin, J.C.-W., Srivastava, G., Guo, D., Zhang, H., Yi, H., Jolfaei, A., 2023. Multi-objective neural evolutionary algorithm for combinatorial optimization problems. IEEE Trans. Neural Networks Learn. Syst. 34 (4), 2133–2143.

She, Y., Li, S., Zhao, Y., 2018. Onboard mission planning for agile satellite using modified mixed-integer linear programming. Aerosp. Sci. Technol. 72, 204–216.

Song, Y., Ou, J., Suganthan, P.N., Pedrycz, W., Yang, Q., Xing, L., 2023a. Learning adaptive genetic algorithm for Earth electromagnetic satellite scheduling. IEEE Trans. Aerosp. Electron. Syst. 59 (6), 9010–9025.

Song, Y., Wei, L., Yang, Q., Wu, J., Xing, L., Chen, Y., 2023b. RL-GA: a reinforcement learning-based genetic algorithm for electromagnetic detection satellite scheduling problem. Swarm Evol. Comput. 77, 101236.

Song, Y., Wu, Y., Guo, Y., Yan, R., Suganthan, P.N., Zhang, Y., Pedrycz, W., Das, S., Mallipeddi, R., Ajani, O.S., et al., 2024. Reinforcement learning-assisted evolutionary algorithm: a survey and research opportunities. Swarm Evol. Comput. 86, 101517.

Song, Y., Ou, J., Pedrycz, W., Suganthan, P.N., Wang, X., Xing, L., Zhang, Y., 2024. Generalized model and deep reinforcement learning-based evolutionary method for multitype satellite observation scheduling. IEEE Trans. Syst., Man, Cybernet.: Syst. 54 (4), 2576–2589.

Sutton, R.S., Barto, A.G., 2018. Reinforcement Learning: An Introduction, 2 edn. MIT Press, Cambridge, MA, USA.

Wang, X., Wu, G., Xing, L., Pedrycz, W., 2020. Agile earth observation satellite scheduling over 20 years: formulations, methods, and future directions. IEEE Syst. J. 15 (3), 3881–3892.

Wang, J., Song, G., Liang, Z., Demeulemeester, E., Hu, X., Liu, J., 2023. Unrelated parallel machine scheduling with multiple time windows: an application to Earth observation satellite scheduling. Comput. Oper. Res. 149, 106010.

Wang, Y., Liu, J., Tong, Y., Yang, Q., Liu, Y., Mou, H., 2023. Resource scheduling in mobile edge computing using improved ant colony algorithm for space information network. Int. J. Satell. Commun. Network. 41 (4), 331–356.

Wang, X., Han, C., Leus, R., 2025. Scheduling multiple agile Earth observation satellites with multiple observations. Adv. Space Res.

Wu, G., Luo, Q., Du, X., Chen, Y., Suganthan, P.N., Wang, X., 2022. Ensemble of metaheuristic and exact algorithm based on the divide-and-conquer framework for multisatellite observation scheduling. IEEE Trans. Aerosp. Electron. Syst. 58 (5), 4396–4408.

Xhafa, F., Sun, J., Barolli, A., Biberaj, A., Barolli, L., 2012. Genetic algorithms for satellite scheduling problems. Mobile Inform. Syst. 8 (4), 351–377.

Xu, B., Rang, G., Xie, R., Li, W., Gong, D., Fan, Z., Yang, S., He, J., 2025. A prediction approach based on long short-term memory networks for dynamic multiobjective optimization. Expert Syst. Appl. 283, 127792.

Xu, B., Zheng, Y., Li, W., Gao, X., Gong, D., He, J., Fan, Z., 2025. Handling multiobjective optimization problems with complex constraints: A constraints grouping-based approach. IEEE Trans. Syst., Man, Cybernet.: Syst.

Zhang, J., Xing, L., 2022. An improved genetic algorithm for the integrated satellite imaging and data transmission scheduling problem. Comput. Oper. Res. 139, 105626.

Zhang, J., Xing, L., Peng, G., Yao, F., Chen, C., 2019. A large-scale multiobjective satellite data transmission scheduling algorithm based on SVM + NSGA-II. Swarm Evol. Comput. 50, 100560.

Zuo, T., Di, K., Li, P., Jiang, Y., 2026. Autonomous domain adaptation self-optimization approach for cross-domain industrial agents. ACM Trans. Intell. Syst. Technol. 17 (1), 1–25.